

Delivery 2 A/B Testing

In this delivery your goal is to evaluate whether a gameplay measure — in this case the strength of a purchasable boost — should be rolled out to all players. You will design and carry out an A/B test analysis, provide arguments for and against increasing the boost's power, and support every conclusion with solid statistical evidence. Beyond the main question, you are also expected to explore the dataset to uncover other relevant insights about the game's performance and player behavior. The ultimate aim is to demonstrate that you can frame an experimental question, analyze player data rigorously, and tell a convincing story based on your findings.

Game & Test Setup

The case study is based on a casual puzzle game similar to *Candy Crush Saga*, where levels are short and aligned linearly. Players must complete one level before unlocking the next, and they can purchase boosts that make levels easier to clear.

On **January 1st, 2020**, the game was launched with all players using the original version of the boost. To address complaints about difficulty, the boost's power was increased for half of the players, forming the **test group (A)**, while the other half remained as the **control group (B)**. This adjustment took place on **June 1st, 2020**. On **October 1st, 2020**, the boost power was reset to its original state for everyone.

Your task is to determine whether the increased boost power should be implemented for all players. To do so, you will analyze KPIs across player groups and demographics, compare the test and control groups, evaluate statistical significance, examine how the effect changes over time, search for unexpected insights ("easter eggs"), and reflect on how the experimental setup could be improved.

Deliverables

1. **Oral Presentation (max 10 minutes):** Present your findings as a coherent story, supported by visuals such as charts or plots. Make sure your arguments for and against are balanced and statistically proven.
2. **Written Report (≈5 pages):** Summarize your approach, methods, findings, and recommendations. The report should include plots, tables, and explanations of statistical tests used.

Evaluation criteria

Trend Discovery & Data Relevance (40%)

- **9–10 (Excellent):** All relevant trends are clearly identified and explained; KPIs are highly relevant and clearly linked to the case question; insights are robust and data-grounded.
 - **7–9 (Good):** Most important trends are identified and explained; KPIs are generally relevant but may lack depth or full connection to the case question; insights are supported by data but some gaps remain.
 - **5–7 (Satisfactory):** Some relevant trends are identified, but important insights are missing; KPI selection is incomplete or inconsistent; links to the case question are weak.
 - **0–4 (Poor):** Trends are missing or irrelevant; KPI selection is unclear or absent; no connection to the case question. KPIs are wrongly calculated.
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Statistical Analysis & Proof (40%)

9–10 (Excellent):

- Correctly applies appropriate statistical methods (e.g., significance testing, effect size calculation).
- Clearly interprets results with explicit reporting of p-values and confidence intervals.
- Considers time-based changes and experimental setup.
- Evidence is strong, statistically sound, and clearly supports conclusions.

7–9 (Good):

- Applies statistical methods mostly correctly; significance tests and confidence values are reported but not always fully explained or consistent.
- Results are explained with some depth; time-based effects considered but not consistently.
- Evidence generally supports conclusions, with minor gaps.

5–7 (Satisfactory):

- Statistical methods are partially correct or incomplete; significance and confidence values are missing or incorrectly reported.

- Explanations of results are limited.
- Time-based effects ignored.
- Evidence is weak or inconsistent.

0–4 (Poor):

- No correct statistical methods applied.
- No significance testing or confidence intervals reported.
- Results are unexplained and unsupported.

Presentation & Communication (20%)

- **9–10 (Excellent):** Findings are clearly and convincingly presented; visuals are highly relevant, well-designed, and support the story; recommendations are strong, clear, and supported by analysis.
- **7–9 (Good):** Findings are understandable and mostly well presented; visuals are relevant but simple; story flow is adequate; recommendations are partially supported.
- **5–7 (Satisfactory):** Findings are unclear or incomplete; visuals are poor or basic; story flow is fragmented; recommendations are weak or poorly justified.
- **0–4 (Poor):** Findings are incoherent or missing; visuals absent or inappropriate; no logical presentation; no recommendations.